Technical Supplement to Certified Model Ranking under Limited Label Errors

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This supplement gives the detailed proofs for the correction lists and the conflict-aware certificate. S1 defines the shared notation and the required approximation. S2 and S3 construct the spectral and sample-mean lists. S4 proves the conflict bound and its lower bound on the breakdown budget. S5 states how to verify a supplied upper bound. S6 proves the exact-decision result and the extension to one fixed replacement label. It also gives the additional examples. S7 describes the exploratory synthetic implementation check and its results. The proofs use Proposition 2 of the main paper for rounding. They use Section IV-D of the main paper to choose a construction and transfer score error to reversal-weight error.

S1 Shared notation and the required approximation

There are $M \geq 2$ fixed binary prediction rows on $N \geq 1$ examples. Only the $r \leq N$ flagged binary labels can change. All other labels stay fixed. The integer error budget satisfies $0 \leq B \leq r$. Write $P_{iu} \in \{0,1\}$ for model i's prediction on example u, and $\widehat{z}_u \in \{0,1\}$ for that example's recorded label. Let $u_1, \ldots, u_r$ be the flagged column indices. A correction specifies the flagged labels:

$$Z_B = \left\{ z \in \{0,1\}^r : \sum_{k=1}^{r} \mathbf{1}\{z_k \neq \widehat{z}_{u_k}\} \leq B \right\}.$$

For $z \in Z_B$, let $\widetilde{z} \in \{0,1\}^N$ be its extension to all examples: $\widetilde{z}_{u_k} = z_k$ on flagged columns and $\widetilde{z}_u = \widehat{z}_u$ elsewhere. Define $c_i(z) = \sum_{u=1}^{N} \mathbf{1}\{P_{iu} = \widetilde{z}_u\}$. The associated flip vector has $x_k = \mathbf{1}\{z_k \neq \widehat{z}_{u_k}\}$. Its coordinate index refers to its position in the ordered flagged set. The feasible binary vectors and the corresponding fractional set are

$$X_B = \{x \in \{0,1\}^r : \textstyle\sum_u x_u \leq B\}, \qquad K_B = \{x \in [0,1]^r : \textstyle\sum_u x_u \leq B\}.$$

The fractional set allows coordinates between zero and one. Their sum still satisfies the same error budget. Every point of K_B satisfies $\|x\|_2^2 \leq \sum_u x_u \leq B$. A reference order gives the direction of each comparison $e = (j, i)$. The comparison tests the later row i against the earlier row j. The rational comparison weights $\pi_e \geq 0$ sum to one. For $e = (j, i)$, let $d_e(z) = (c_i(z) - c_j(z))/N$. For a margin t, define

$$C_{\pi,t}(z) = \sum_e \pi_e \mathbf{1}\{d_e(z) > t\}, \qquad U_B(t) = \max_{z \in Z_B} C_{\pi,t}(z).$$

For rational $t \geq 0$ and $0 \leq \tau < 1$, the audit asks whether $U_B(t) \leq \tau$. Equality at the margin does not count as a reversal. The notation $d_e(x)$ denotes the same score in the associated binary flip coordinates. Its affine extension to fractional vectors is $d_e(x) = \beta_e + a_e^T x/N$, where β_e is its recorded difference. The coordinates of a_e follow the same ordered flagged set. A binary flip changes each

row's correct count by 1 or -1, so $a_{eu} \in \{-2, 0, 2\}$. The original denominator N does not change. Set

$$\Sigma = \sum_e \pi_e a_e a_e^T, \qquad \|v\|_{\Sigma,N}^2 = \frac{v^T \Sigma v}{N^2} = \sum_e \pi_e \left(\frac{a_e^T v}{N}\right)^2. \tag{S1}$$

This squared seminorm measures the weighted sum of squared changes in pair scores. The matrix satisfies $\Sigma \succeq 0$, $\Sigma_{uu} \leq 4$, and $\operatorname{tr} \Sigma \leq 4r$. A nonzero vector can have zero seminorm. Thus the proofs do not require an invertible matrix. The input length ℓ counts the predictions, flagged columns, recorded and fixed labels, reference order, rational weights, and numerical parameters. These objects are listed explicitly. Their dimensions are therefore bounded by a polynomial in ℓ.[1]

For rational $\delta, \epsilon \in (0, 1)$, let $j \geq 0$ be the least integer with $4^{-j} \leq \epsilon$ and define

$$\nu = \delta 2^{-j}, \qquad \epsilon\delta^2/4 < \nu^2 \leq \epsilon\delta^2. \tag{S2}$$

Exact rational comparisons find j, with polynomial input bit cost. Each construction returns a nonempty list of feasible corrections. The list is a cover if every feasible correction has a nearby representative in the list:

$$\forall x \in X_B \; \exists w \in W \subseteq X_B : \quad \|x - w\|_{\Sigma,N} \leq \nu. \tag{S3}$$

Proposition 2 of the main paper gives polynomial-bit rational rounding from any $y \in K_B$ to $w \in X_B$, with $\|w - y\|_{\Sigma,N}^2 \leq 8B/N^2$. Section IV-D of the main paper gives the integer rule for choosing a construction. It also converts the cover error into bounds that hold for every margin and preserve strict reversals. When $\delta \geq 2B/N$, the recorded-label correction alone gives those bounds. This direct case need not satisfy (S3).

S2 A rational spectral cover

The spectral construction separates directions with large score variation from directions with small variation. The trace bound limits the number of directions that must be retained. The algorithm builds their subspace one rational column at a time. An exact test either bounds the variation that remains or gives the next direction to retain. A grid groups corrections with similar retained scores. Each nonempty grid cell gives a feasible fractional representative. Rounding then gives a binary correction under the same error budget. The proof bounds each source of error and counts all grid tests. All stored quantities are rational.[2] The method of retaining covariance directions and controlling squared error follows Diakonikolas et al. [S3, Sec. 2.1, Lemma 2.2 and Algorithm 1]. The proof below gives a rational construction under the binary error budget. It includes the subspace, the grid search inside an ellipsoid, the feasible representatives, and the bit bounds.

Theorem S2.1 (Rational spectral cover)**.** *Let the explicit rational input and X_B, K_B, Σ, and ν be as in Section S1. A deterministic algorithm returns a nonempty list $W \subseteq X_B$ that satisfies* (S3)*. Its size and bit-operation count are at most*

$$2^{O(1+Br/(\epsilon\delta^2 N^2))} \operatorname{poly}(\ell),$$

where ℓ is the explicit rational input length. The polynomial degree is an absolute constant. Each output gives a feasible binary correction.

[1] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

[2] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

Proof. The shared coefficient bound gives $\operatorname{tr}\Sigma \le 4r$ and $\Sigma \succeq 0$. Also $\|x\|_2^2 \le B$ for $x \in K_B$. Put

$$E = \frac{Br}{N^2}, \qquad \rho = \frac{E}{\nu^2}.$$

If $B = 0$, $r = 0$, or $\Sigma = 0$, output only the zero flip vector. If $\rho \le 1/4$, do the same. In this case, for every $x \in X_B$,

$$\|x\|_{\Sigma,N}^2 \le \frac{\operatorname{tr}\Sigma\,\|x\|_2^2}{N^2} \le 4E \le \nu^2.$$

Thus these cases satisfy the cover condition.

Assume $\rho > 1/4$. If $N^2 < 128B/\nu^2$, enumerate all 2^r binary vectors and retain those in X_B. The list contains the zero vector and every other feasible vector. Since $r < 128\rho$, this enumeration has the stated cost. For the rest of the proof, assume

$$N^2 \ge 128B/\nu^2. \tag{S4}$$

Proposition 2 of the main paper rounds any rational point $y \in K_B$ to a feasible $w \in X_B$ using a polynomial number of bit operations with $(w-y)^T\Sigma(w-y) \le 8B$. By (S4), this rounding has score error at most $\nu/4$.

Rational retained subspace. A grid will cover the retained part. We first bound the score error in the remaining part for every feasible correction. Set $C = 4r$ and $\lambda = \nu^2N^2/(128B)$. We construct a rational orthogonal projector Π such that

$$R = (I-\Pi)\Sigma(I-\Pi) \preceq 2\lambda I, \qquad k = \operatorname{rank}\Pi \le \lceil 512\rho \rceil \le 516\rho. \tag{S5}$$

Start with $\Pi = 0$. At each step, use exact symmetric elimination to test whether $H = 2\lambda I - R$ is positive semidefinite. A positive diagonal pivot is removed by a rational Schur complement. A negative diagonal gives a rational direction with a negative quadratic form. If a zero diagonal has a nonzero off-diagonal entry, a rational two-coordinate vector with a sufficiently small coordinate of the opposite sign has a negative quadratic form. If all remaining entries are zero, the test passes. If a residual block gives a negative direction, back substitution gives a rational vector $v \ne 0$ with $v^TRv > 2\lambda v^Tv$. Fraction-free elimination gives polynomial bit length for the test and its witness. The equivalent ratios of minors give the same bound.

Let $v_0 = v/\|v\|_2$ only for analysis. Define

$$\eta = \min\{1/8, \lambda/(16C)\}, \qquad D = \lceil 4r/\eta \rceil.$$

Approximate each normalized coordinate using rational bisection and squared comparisons with v_i^2 and v^Tv. Place each approximation to $|v_{0i}|$ on the grid $D^{-1}\mathbb{Z}$. Restore its sign from the rational vector. This gives $q \in D^{-1}\mathbb{Z}^r \cap [-1,1]^r$ with $\|q - v_0\|_2 \le \sqrt{r}/D \le \eta$. No normalized irrational vector is stored. Put $u = (I-\Pi)q$. Since $R \succeq 0$ and $\|R\|_{\mathrm{op}} \le C$,

$$q^TRq > 2\lambda - C\eta(2+\eta), \qquad \|u\|_2 \le 1+\eta.$$

Here $u^T\Sigma u = q^TRq$. The numerator is at least $(239/128)\lambda$, and $(1+\eta)^2 \le 81/64$. Therefore $u^T\Sigma u/u^Tu > \lambda$, so $u \ne 0$. Add q, or equivalently u, to the stored subspace. The residual trace decreases by $u^T\Sigma u/u^Tu > \lambda$. Its initial value is at most $4r$ and it stays nonnegative. Fewer than $4r/\lambda = 512\rho$ additions occur. The process stops only when $H \succeq 0$, which proves (S5).

Store the original grid columns in V. This avoids recursively expanding the residual vectors. They are linearly independent because each new u is nonzero. If $V = V_{\mathbb{Z}}/D$, then

$$\Pi = V(V^TV)^{-1}V^T = V_{\mathbb{Z}}(V_{\mathbb{Z}}^TV_{\mathbb{Z}})^{-1}V_{\mathbb{Z}}^T. \tag{S6}$$

Each integer Gram entry has magnitude at most rD^2. Hadamard's determinant bound gives polynomial bit length for every Gram minor, the inverse, and Π. The same bound applies when exact rational operations make these columns orthogonal.

A rational score map and its enclosing ellipsoid. If $k > 0$, rational Gram–Schmidt gives orthogonal columns $V_\perp = (v_1, \ldots, v_k)$ spanning the stored subspace. Gram inverses express each column as a rational combination of the stored columns. The determinant bound above controls its bit length. Set

$$G = V_\perp^TV_\perp, \qquad J = V_\perp^T\Sigma V_\perp.$$

The matrix G is positive diagonal, $J \succeq 0$, and $\Pi = V_\perp G^{-1}V_\perp^T$. Factor J in the fixed coordinate order as $J = L\,\mathrm{diag}(d_i)L^T$, with unit lower-triangular rational L. A positive pivot uses its Schur complement. If a residual pivot is zero, positive semidefiniteness makes its residual row and column zero, since each relevant 2×2 principal minor is $-J'^2_{ij} \geq 0$. Set that $d_i = 0$ and set its new multipliers to zero. Keep the multipliers from earlier positive pivots. Thus a zero pivot preserves the original row of J.

Powers of two let the algorithm work without storing pivot square roots. For $d_i > 0$, choose the least integer e_i with $2^{2e_i} \geq d_i$, and put $s_i = 2^{e_i}$. Put $s_i = 0$ when $d_i = 0$, and define $S = \mathrm{diag}(s_i)L^T$. Exact squared-power comparisons give

$$J \preceq S^TS \preceq 4J. \tag{S7}$$

The inequality also holds for singular J and singular S. To bound the bit length, clear the input denominators. Every Schur entry equals a bordered minor divided by a positive principal minor on the earlier positive pivots. Each multiplier is a ratio of these entries. Determinant bounds give polynomial bit length. This also holds when a positive pivot is small. A pivot with b bits has $|e_i| = O(b)$, so S has polynomial bit length. No inverse of J or S is used.

Define the rational map and shapes

$$\zeta(x) = \frac{1}{N}SG^{-1}V_\perp^Tx, \qquad Q = \frac{B}{N^2}SG^{-1}S^T, \qquad Q_+ = Q + \nu^2I_k. \tag{S8}$$

Equation (S7) gives

$$\|\Pi(x-y)\|_{\Sigma,N} \leq \|\zeta(x) - \zeta(y)\|_2. \tag{S9}$$

For $x \in K_B$, the vector $a = G^{-1/2}V_\perp^Tx/\sqrt{B}$ has norm $\|\Pi x\|_2/\sqrt{B} \leq 1$. In this analysis only, put $F = (\sqrt{B}/N)SG^{-1/2}$. Then $\zeta(x) = Fa$ and $Q = FF^T$. Since $Q \preceq Q_+$, the operator norm of $Q_+^{-1/2}F$ is at most one. Hence $\zeta(K_B)$ lies in $\mathcal{E}(Q_+) = \{z : z^TQ_+^{-1}z \leq 1\}$, even if Q is singular. The square roots are used only to prove this inclusion. The algorithm computes the rational matrices in (S8). These matrices also satisfy

$$Q_+ \succeq \nu^2I_k, \qquad \mathrm{tr}\,Q \leq \frac{4B}{N^2}\,\mathrm{tr}(\Pi\Sigma) \leq 16E, \qquad \mathrm{tr}\,Q_+ \leq 532E. \tag{S10}$$

The last bound uses $k\nu^2 \leq 516E$.

Exact grid search one coordinate at a time. The search builds each integer grid index one coordinate at a time. Its first coordinates form a prefix. A prefix is rejected only if no real values of the remaining coordinates can satisfy the ellipsoid inequality. For $k > 0$, choose the rational mesh, dyadic scale, and range

$$h = \frac{\nu}{16\lceil\sqrt{k}\rceil}, \qquad \sqrt{E} \le g < 2\sqrt{E}, \qquad m_{\max} = \lfloor 48g/h \rfloor.$$

Use exact squared comparisons to find $\lceil\sqrt{k}\rceil$ and the power of two g. Use closed cubes of side h centered at hm for the integer vectors m with

$$h^2 m^T Q_+^{-1} m \le 4. \tag{S11}$$

For $x \in K_B$, round each coordinate of $\zeta(x)$ to a nearest multiple of h. The Euclidean error is at most $h\sqrt{k}/2 \le \nu/32$. By (S10), its Q_+^{-1}-norm is at most $1/32$. Thus the center is in $(33/32)\mathcal{E}(Q_+)$ and satisfies (S11). The closed cube around this center contains $\zeta(x)$. This also holds when rounding gives a boundary tie.

Let $Q_{+,j}$ be the leading $j \times j$ block. For $j < k$, write $Q_+ = \left(\begin{smallmatrix} A & T \\ T^T & D_0 \end{smallmatrix}\right)$, where $A = Q_{+,j}$ and $R_0 = D_0 - T^T A^{-1} T \succ 0$. Block inversion gives

$$(p, v)^T Q_+^{-1} (p, v) = p^T A^{-1} p + (v - T^T A^{-1} p)^T R_0^{-1} (v - T^T A^{-1} p). \tag{S12}$$

Its minimum over real extensions v is $p^T Q_{+,j}^{-1} p$. Start with the empty prefix. Below each retained prefix of length $j-1$, try every next integer coordinate in $[-m_{\max}, m_{\max}]$. Retain the new prefix exactly when

$$h^2 m_{[j]}^T Q_{+,j}^{-1} m_{[j]} \le 4. \tag{S13}$$

Use exact rational comparisons and retain equality. Each coordinate of a center in $2\mathcal{E}(Q_+)$ has magnitude at most $2\sqrt{(Q_+)_{ii}} \le 2\sqrt{532E} < 48g$. Thus the finite coordinate range includes every required center. Equation (S12) proves that every accepted full center has accepted prefixes. Induction on j proves that the traversal returns exactly the centers in (S11). A retained prefix can still have no integer continuation.

Let n_j be the number of retained prefixes at depth j, with $n_0 = 1$. Their side-h cubes have disjoint interiors. Each lies in $3\mathcal{E}(Q_{+,j})$ because its Euclidean radius is at most $\nu/32$ and $Q_{+,j} \succeq \nu^2 I_j$. Comparing their total volume with the ellipsoid volume gives

$$n_j h^j \le 3^j \operatorname{vol}(\mathbb{B}_2^j) \sqrt{\det Q_{+,j}}, \qquad n_j \le \left(\frac{C_1\sqrt{E}}{jh}\right)^j$$

for an absolute constant C_1. The second bound uses $\det Q_{+,j} \le (532E/j)^j$ and $\operatorname{vol}(\mathbb{B}_2^j) \le (C_2/\sqrt{j})^j$. Since $\sqrt{E}/h = 16\sqrt{\rho}\lceil\sqrt{k}\rceil = O(\rho)$ and $u\log(b/u) \le b/e$, every n_j is $\exp(O(\rho))$. Also $m_{\max} = O(\rho)$ and $k = O(\rho)$. The total number of attempted children is therefore

$$(2m_{\max} + 1) \sum_{j=0}^{k-1} n_j = 2^{O(\rho)}. \tag{S14}$$

This count includes rejected children and retained prefixes that have no integer continuation. Thus the search cost does not require scanning the full coordinate box.

Feasible representatives and feasible outputs. For every retained full cell, test whether the following rational linear constraints have a solution:

$$y \in K_B, \qquad hm_i - h/2 \le \zeta_i(y) \le hm_i + h/2 \quad (1 \le i \le k).$$

An exact rational linear program returns a rational feasible y. If no solution exists, it returns a rational Farkas certificate that proves the cell is empty. Keep one representative for each nonempty cell. Every $x \in X_B$ lies in such a cell. For its representative y, (S9) bounds the retained-part error by $h\sqrt{k} \le \nu/16$. Also $\|x - y\|_2 \le 2\sqrt{B}$ and (S5) give

$$\|(I - \Pi)(x - y)\|_{\Sigma,N} \le \frac{\sqrt{8\lambda B}}{N} = \nu/4.$$

The retained part and the remaining part need not be orthogonal in the score seminorm. The triangle inequality gives $\|x - y\|_{\Sigma,N} \le 5\nu/16$. Apply Proposition 2 of the main paper to each representative. Its feasible output w has $\|y - w\|_{\Sigma,N} \le \nu/4$ by (S4). Therefore $\|x - w\|_{\Sigma,N} \le 9\nu/16 < \nu$.

If $k = 0$, use one formal cell with no coordinate constraints and representative $y = 0$. The residual bound gives a score error at most $\nu/4$ for any $x \in X_B$, and rounding preserves $y = 0$. The list is nonempty in all branches because $0 \in X_B$. Duplicate rounded vectors can be removed after the output for each nonempty cell is recorded. All outputs satisfy the exact budget.

Bit cost. The scalars E, ρ, ν, and λ have polynomial input bit length. Here $D = O(r \max\{1, \rho\})$. The Gram and minor bounds above give polynomial bit length in ℓ, r, k, and $\log(2 + \rho)$ for the projector, orthogonal basis, factor, and shape. The exponent of dyadic g is bounded by the bit length of E. The mesh, coordinate range, principal inverses, and cell inequalities have the same type of bound. In particular, a small positive E increases the bit length. It does not require $1/E$ iterations. Each integer cell index has $O(\log(2 + \rho))$ bits.

Each cell linear program has r variables and $O(r + k)$ rational constraints. Standard exact rational linear programming returns a feasible point of polynomial bit length, or a polynomial-bit Farkas certificate, using a polynomial number of bit operations. Proposition 2 of the main paper gives the rounding bit bound. Each step outside enumeration has fixed-degree polynomial cost in ℓ and polynomial cost in k and ρ. The factor $2^{O(\rho)}$ absorbs these parameter polynomials because $\rho > 1/4$. Equation (S14) bounds the full traversal and all cell tests. Finally, (S2) gives $\rho < 4Br/(\epsilon\delta^2 N^2)$. This proves the stated list size and bit-operation bound. The main paper then converts the cover error into the reversal-weight bound. That argument holds for every real margin and preserves strict reversals. □

S3 The sample-mean construction

The sample-mean construction has three steps. First, the algorithm enumerates means that approximate every point of K_B in score seminorm. A mean can have a coordinate above one. Second, conditional-gradient (Frank–Wolfe) steps use exact rational arithmetic to minimize distance to the mean up to a stated error. Their output is a feasible fractional point. Third, Proposition 2 of the main paper rounds the point to a feasible binary vector. A probability argument proves that this deterministic list covers every correction. It does not assign a probability model to the test labels.[3]

[3] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

Sparse convex approximation and Frank–Wolfe are established tools [S5, S6]. The proof gives their error and bit bounds under the same error budget.

Theorem S3.1 (Sample-mean cover)**.** *Under the assumptions of Section S1, a deterministic algorithm constructs a nonempty $W_X \subseteq X_B$ satisfying* (S3)*. Its size and bit-operation count are at most*

$$(r+1)^{O(1+B^2/(\epsilon\delta^2N^2))}\,\mathrm{poly}(\ell). \tag{S15}$$

The polynomial degree is an absolute constant, independent of B, δ, ϵ.

Proof. If $B = 0$ or $r = 0$, return the zero vector, which is the only member of X_B. If $\Sigma = 0$, return zero because every score distance vanishes. These cases have polynomial bit cost. For the rest of the proof, assume $1 \le B \le r$ and $\Sigma \neq 0$. Compute

$$k = \left\lceil \frac{64B^2}{\nu^2N^2} \right\rceil, \qquad T = \left\lceil \frac{512Br}{\nu^2N^2} \right\rceil \tag{S16}$$

by exact rational arithmetic. Both integers are positive.

Fallback by feasible subset enumeration. If $N^2 < 128B/\nu^2$, enumerate all subsets of $\{1, \dots, r\}$ of size at most B. Start with the empty subset. Append each larger index while the size is less than B, and output the incidence vector at every node. Each feasible subset has exactly one path with increasing indices. Thus the output is nonempty and covers X_B exactly. Pad each increasing list with zeros to length B. Distinct lists give distinct words over $\{0, \dots, r\}$. Therefore

$$|X_B| = \sum_{j=0}^{B} \binom{r}{j} \le (r+1)^B, \qquad \frac{64B^2}{\nu^2N^2} > \frac{B}{2}, \qquad B < 2k.$$

The traversal and each output require polynomial work per subset. This branch costs at most $(r+1)^{2k}\,\mathrm{poly}(\ell)$. For the remaining construction, assume

$$N^2 \ge 128B/\nu^2. \tag{S17}$$

A finite set of means. Let e_u be the uth coordinate unit vector and set $\mathcal{A} = \{0, Be_1, \dots, Be_r\}$. Enumerate all $(r+1)^k$ ordered words in $\mathcal{A}^k$ in lexicographic order. Keep the rational mean $c = k^{-1}\sum_{i=1}^{k} Y_i$ for every word, including repeated means.

To prove coverage, fix $x \in K_B$. For this proof, choose a random vector from the same finite set with probabilities

$$\Pr(Y = Be_u) = x_u/B, \qquad \Pr(Y = 0) = 1 - \sum_u x_u/B.$$

These probabilities are nonnegative, sum to one, and give $\mathbb{E}Y = x$. Expand the squared seminorm around this mean to obtain

$$\mathbb{E}\|Y - x\|_{\Sigma,N}^2 = \mathbb{E}\|Y\|_{\Sigma,N}^2 - \|x\|_{\Sigma,N}^2 \le \frac{B}{N^2}\sum_u x_u\Sigma_{uu} \le \frac{4B^2}{N^2}.$$

For k independent copies, the centered cross terms have zero expectation. Averaging therefore gives

$$\mathbb{E}\left\| \frac{1}{k}\sum_{i=1}^{k} Y_i - x \right\|_{\Sigma,N}^2 \le \frac{4B^2}{kN^2} \le \frac{\nu^2}{16}.$$

At least one word has squared error at most this expectation. The algorithm enumerates that word. Hence

$$\forall x \in K_B \; \exists c: \quad \|c - x\|_{\Sigma,N} \le \nu/4. \tag{S18}$$

Thus the deterministic list contains a suitable mean. Every mean has nonnegative coordinates with sum at most B. A coordinate can still exceed one. Before rounding, we must replace the mean by a point in K_B.

Feasible fractional points. The next step enforces the coordinate bounds. For each mean c, approximately minimize the convex quadratic $f_c(y) = \|y - c\|^2_{\Sigma,N}$ over K_B as follows. Start at $y_0 = 0$. At step s, form $g_s = 2\Sigma(y_s - c)/N^2$. Order its negative coordinates from smallest to largest, with ties broken by coordinate index. Let v_s have ones at the first at most B such coordinates and zeros elsewhere. Then $v_s \in X_B$ minimizes $g_s^T v$ over K_B. Assigning mass to a positive coefficient cannot decrease the objective. Zero coefficients need no mass. Moving mass from a larger negative coefficient to a smaller one cannot increase the objective. If the budget is not full, adding mass at an unfilled negative coordinate decreases the objective. With the unit bounds and the integer budget, these moves give an optimum that fills precisely the selected coordinates. Thus the linear oracle sorts the negative coefficients and selects at most B of the smallest ones. It leaves positive and zero coefficients unused. Use T updates

$$\gamma_s = \frac{2}{s+2}, \qquad y_{s+1} = (1 - \gamma_s)y_s + \gamma_s v_s, \qquad 0 \le s < T. \tag{S19}$$

Convexity keeps every iterate in K_B.

For $u, v \in K_B$, nonnegativity gives $\|u - v\|_2^2 \le \|u\|_2^2 + \|v\|_2^2 \le 2B$. Since $\|\Sigma\|_{\mathrm{op}} \le \operatorname{tr}\Sigma \le 4r$,

$$\|u - v\|^2_{\Sigma,N} \le \frac{8Br}{N^2} =: D_0. \tag{S20}$$

The feasible set is compact, so a minimizer y^* exists. Write its value as f_c^*. Put $h_s = f_c(y_s) - f_c^* \ge 0$. Convexity and the linear minimum give

$$\langle g_s, v_s - y_s \rangle \le \langle g_s, y^* - y_s \rangle \le -h_s.$$

The exact quadratic expansion and (S20) give

$$\begin{aligned} h_{s+1} &= h_s + \gamma_s \langle g_s, v_s - y_s \rangle + \gamma_s^2 \|v_s - y_s\|^2_{\Sigma,N} \\ &\le (1 - \gamma_s)h_s + \gamma_s^2 D_0. \end{aligned} \tag{S21}$$

Since $\gamma_0 = 1$, we get $h_1 \le D_0$. This does not depend on h_0 or the location of c. We claim $h_s \le 4D_0/(s+2)$ for $s \ge 1$. The first case follows from $D_0 \le 4D_0/3$. If the bound holds at s, then

$$h_{s+1} \le \frac{4D_0(s+1)}{(s+2)^2} \le \frac{4D_0}{s+3},$$

where $(s+1)(s+3) \le (s+2)^2$. Induction and (S16) now give

$$f_c(y_T) - f_c^* \le \frac{32Br}{N^2(T+2)} \le \frac{\nu^2}{16}. \tag{S22}$$

For the mean that covers a given $x \in K_B$ in (S18), feasibility of x gives $f_c^* \le f_c(x) \le \nu^2/16$. Consequently

$$\|y_T - c\|_{\Sigma,N} \le \nu/\sqrt{8}. \tag{S23}$$

We construct y_T for every word. Thus the algorithm need not identify which word covers a given x. The argument also allows Σ to be singular.

Feasible binary outputs and their error. Apply Proposition 2 of the main paper to each rational y_T. It gives $w \in X_B$ with

$$\|w - y_T\|_{\Sigma,N}^2 \le \frac{8B}{N^2} \le \frac{\nu^2}{16},$$

where the last bound uses (S17). Keep an output for every word. In particular, the all-zero word ensures a nonempty output list. For each $x \in X_B$, choose a mean from (S18). The seminorm triangle inequality gives

$$\begin{aligned} \|x - w\|_{\Sigma,N} &\le \|x - c\|_{\Sigma,N} + \|c - y_T\|_{\Sigma,N} + \|y_T - w\|_{\Sigma,N} \\ &\le \left(\frac{1}{2} + \frac{1}{\sqrt{8}}\right)\nu < \nu. \end{aligned} \tag{S24}$$

This proves (S3), including singular Σ. Duplicate outputs may be kept. If they are removed, the coverage record must still identify the output for every word.

Exact arithmetic and total cost. Let Q_π be the product of the denominators of the rational weights. Its bit length is at most the total input bit length of those denominators. The matrix $Q_\pi\Sigma$ is integral, with entries of absolute value at most $4Q_\pi$, and has polynomial construction cost. A mean has coordinates Bn_u/k, where $0 \le n_u \le k$ and $\sum_u n_u \le k$. Its bit length is polynomial in $\ell + \log(k+1)$.

For every $s \ge 1$, the iterates have the common-denominator formula

$$y_s = \frac{2}{s(s+1)} \sum_{q=0}^{s-1} (q+1) v_q. \tag{S25}$$

At $s = 1$ it gives $y_1 = v_0$. Substitution in (S19) proves the next case. Store the integer sum in this formula. After multiplication by two, each coordinate numerator is between zero and $s(s+1)$. For $s \ge 1$, a common positive gradient denominator is $Q_\pi N^2 k s(s+1)$. At $s = 0$, use $Q_\pi N^2 k$. The bounds $|\Sigma_{uv}| \le 4$, $0 \le (y_s)_u \le 1$, and $0 \le c_u \le B \le r$ give $|(g_s)_u| \le 8r(B+1)/N^2$. Thus gradient numerators have $O(\ell + \log(k+1) + \log(s+1))$ bits. Matrix products, sorting, comparisons, and integer-sum updates have polynomial bit cost in these lengths. The degree is fixed. The final point has denominator $T(T+1)$, so the polynomial rational bit bound in Proposition 2 of the main paper applies to its rounding.

The total cost includes every conditional-gradient iteration. Equation (S16) gives

$$T \le 8(r/B)k + 1 \le 8rk + 1. \tag{S26}$$

Since $r \le \ell$ for explicit input, constructing a word and its mean, performing all T iterations, rounding, and writing the output cost $\mathrm{poly}(\ell, k)$ bit operations per word. The degree is absolute. For $r, k \ge 1$, any fixed polynomial in k can be included in the factor $(r+1)^{O(k)}$. The complete word enumeration therefore costs $(r+1)^{O(k)}\,\mathrm{poly}(\ell)$. The fallback branch has this same form. Finally, (S2) implies

$$k < 1 + \frac{256B^2}{\epsilon\delta^2 N^2},$$

which proves (S15) for size and bit cost. These counts include every word and iteration even when k and T are large. Small rational tolerances also affect the bit length of each operation. □

S4 Conflict-aware certificates and the breakdown budget

Each correction determines which comparisons reverse. The conflict-aware certificate first tests whether each reversal is possible. It then tests whether two reversals can occur together. These tests define a graph. Each edge joins reversals that cannot occur together. The graph constraints hold for every correction. Allowing fractional event indicators gives an upper bound on total reversal weight. A dual solution, defined below, verifies the bound without listing all corrections.[4] Each flagged column has its recorded binary label and its flipped label. Every other column has one fixed label. The reference order, rational weights π_e, and original denominator N are fixed. The error budget B limits the number of flagged labels that change. The edge relaxation is a standard graph bound [S2, S7].

S4.1 Exact event tests

For a reference-ordered pair $e = (j, i)$, define the contribution of label v in column u by

$$g_{eu}(v) = \mathbf{1}\{P_{iu} = v\} - \mathbf{1}\{P_{ju} = v\} \in \{-1, 0, 1\}.$$

Thus $\Delta_e(z) = c_i(z) - c_j(z) = \sum_{u=1}^{N} g_{eu}(\widetilde{z}_u)$. For rational t, put $K = \lfloor Nt \rfloor + 1$. The strict reversal $d_e(z) > t$ is exactly $\Delta_e(z) \geq K$. Write $R_e^B = \{z \in Z_B : \Delta_e(z) \geq K\}$ and let $E_+ = \{e : \pi_e > 0\}$. There are $P = |E_+| \leq \binom{M}{2}$ such events.

Faster binary event tests. In flip coordinates, $\Delta_e(x) = N\beta_e + a_e^T x$. In particular, $\Delta_e(0) = N\beta_e$ is the recorded count difference; the zero argument denotes no label changes. For one binary event, let n_+ count flagged columns with $a_{eu} = 2$. Its largest count difference is $\Delta_e(0) + 2\min\{B, n_+\}$. Thus one scan tests whether the event can reach K.

For two events e, f, count columns by their normalized gain $(a_{eu}/2, a_{fu}/2) \in \{-1, 0, 1\}^2$. Let n_{ab} denote the count of type (a, b) and set

$$A_0 = \left\lceil \frac{K - \Delta_e(0)}{2} \right\rceil, \qquad C_0 = \left\lceil \frac{K - \Delta_f(0)}{2} \right\rceil.$$

A selected set of flips must give normalized gains at least A_0, C_0. Such a set exists if and only if the following test succeeds. Put

$$q = \min\{B, n_{11}\}, \quad b = B - q, \quad A = A_0 - q, \quad C = C_0 - q.$$

For some integer $p \in \{0, \ldots, \min(b, n_{1,-1})\}$, test

$$u = \max\{0, A - p\} \leq n_{10}, \qquad v = \max\{0, C + p\} \leq n_{01}, \qquad p + u + v \leq b. \tag{S27}$$

Also test each $p \in \{0, \ldots, \min(b, n_{-1,1})\}$ with

$$u = \max\{0, A + p\} \leq n_{10}, \qquad v = \max\{0, C - p\} \leq n_{01}, \qquad p + u + v \leq b. \tag{S28}$$

To prove necessity, start with a feasible selected set. If fewer than q columns of type $(1, 1)$ are selected, either add one using a free budget unit or exchange it for another selected type. Each operation weakly increases both gains. Repeat until q such columns are selected. Discard

[4] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

types with both gains nonpositive. Apart from the selected $(1,1)$ columns, the remaining types are $(1,0),(0,1),(1,-1),(-1,1)$. If both tradeoff types occur, remove one of each. Their sum is zero, so neither gain changes. Repeat until at most one tradeoff type remains. If its count is p, the remaining gain requirements give the minimum axis counts u, v in (S27) or (S28). Reducing larger axis counts to these minima preserves feasibility. Their capacities and total cost therefore pass the test. For sufficiency, a passing test directly selects q columns of type $(1,1)$, p of the stated tradeoff type, and u, v of the axis types. This selected set meets both gain thresholds and uses at most B flips.

After the recorded count differences are available, this test uses $O(r+B)$ integer arithmetic operations and a fixed number of counters. Since $B \le r$, this is $O(r)$ for $r \ge 1$. The case $r = 0$ uses the recorded count differences directly. The integer lengths are polynomial in the explicit input length. Storing the selected column indices also gives a correction that witnesses feasibility. This test uses binary gains. The following dynamic program also applies to categorical replacements.

General finite candidate sets. A dynamic program tests two events e, f by processing one column at a time. It records both count differences and the number of changes. Let $\mathcal{D}_u$ be the explicitly listed finite set of permitted labels for column u. It contains the recorded label. For an unflagged column, it contains only that label. In this extension, Z_B is the set of vectors $z \in \prod_{k=1}^{r} \mathcal{D}_{u_k}$ with at most B coordinates that differ from their recorded labels. The extension $\widetilde{z}$ keeps all unflagged labels fixed. The definitions of correct counts, reversal events, and reversal weight use these permitted labels. Put $q_u(v) = \mathbf{1}\{u \text{ is flagged and } v \ne \widehat{z}_u\}$. Start with $T_0 = \{(0,0,0)\}$ and, for $u = 1, \dots, N$, form

$$T_u = \{(a + g_{eu}(v), d + g_{fu}(v), b + q_u(v)) : (a,d,b) \in T_{u-1},\ v \in \mathcal{D}_u,\ b + q_u(v) \le B\}. \tag{S29}$$

The first two coordinates are partial count differences. The third coordinate is the number of changes. At $u = 0$, the single state has this meaning. Suppose the states at $u-1$ have this meaning. Each transition adds one permitted label and gives the exact new counts. Conversely, every permitted partial labeling on u columns has a predecessor on $u-1$ columns and its last label gives a transition. This proves the state correspondence in both directions by induction. The change costs are nonnegative. Thus a state removed for exceeding B cannot later give a feasible correction. Consequently, $R_e^B \cap R_f^B \ne \varnothing$ exactly when T_N contains (a,d,b) with $a, d \ge K$. Omitting the second count gives the exact one-event test.

At layer u, both count coordinates lie in $[-u, u]$, and $b \in \{0, \dots, B\}$. Put $D = \sum_u |\mathcal{D}_u|$, with $D \le 2N$ for binary labels. Each layer uses $O(N^2(B+1))$ Boolean cells. Storing only the previous and current layers keeps total working memory at $O(N^2(B+1))$ cells. The test uses $O(N^2(B+1)D)$ transitions, including table initialization. The one-event test uses $O(N(B+1)D)$ transitions and $O(N(B+1))$ working cells. State indices have $O(\log(N+1))$ bits because $B \le r \le N$. The exact floor of rational Nt, label comparisons, and all other arithmetic have polynomial bit cost in the explicit input length ℓ. This also covers thresholds outside the attainable count range.

S4.2 The graph bound and a sparse dual

Use the one-event test to obtain $V_B = \{e \in E_+ : R_e^B \ne \varnothing\}$. Join two distinct vertices when the two-event test finds $R_e^B \cap R_f^B = \varnothing$; call the edge set F_B. Define

$$H_B(t) = \max\left\{ \sum_{e \in V_B} \pi_e v_e : 0 \le v_e \le 1, \quad v_e + v_f \le 1 \quad (\{e,f\} \in F_B) \right\}. \tag{S30}$$

If V_B is empty, the value is zero. Each $z \in Z_B$ gives a feasible vector with $v_e = \mathbf{1}\{z \in R_e^B\}$. Its objective is $C_{\pi,t}(z)$. Removing all edge constraints leaves the per-comparison bound $B_{\mathrm{sep},B}(t) = \sum_{e \in V_B} \pi_e$. Hence

$$U_B(t) \le H_B(t) \le B_{\mathrm{sep},B}(t). \tag{S31}$$

If an edge joins vertices of weights $\pi_e, \pi_f > 0$, its constraint also gives $H_B(t) \le B_{\mathrm{sep},B}(t) - \min\{\pi_e, \pi_f\}$: if $\pi_e \ge \pi_f$, then $\pi_e v_e + \pi_f v_f = (\pi_e - \pi_f)v_e + \pi_f(v_e + v_f) \le \pi_e$. The graph takes at most P one-event and $\binom{P}{2}$ two-event tests. The dynamic programs use $O(P^2N^2(B+1)D)$ transitions. One rational linear-program solve then gives the bound. Together these steps use a polynomial number of bit operations.

The dual of (S30) is

$$\begin{aligned} \min \quad & \sum_{e \in V_B} \alpha_e + \sum_{\{e,f\} \in F_B} \beta_{ef}, \\ \text{subject to} \quad & \alpha_e + \sum_{f : \{e,f\} \in F_B} \beta_{ef} \ge \pi_e \quad (e \in V_B), \\ & \alpha_e, \beta_{ef} \ge 0. \end{aligned} \tag{S32}$$

It is feasible with $\alpha_e = \pi_e$ and $\beta = 0$. The primal is feasible and bounded. Rational linear-program duality gives an attained common optimum. For $n = |V_B| > 0$, add nonnegative surplus variables to make the n covering constraints equalities. The matrix has full row rank because it contains the n identity columns for α. Its entries are in $\{-1, 0, 1\}$. An optimal basic solution therefore has at most n positive variables, including surplus variables. Thus it has at most n positive α and β entries. A sparse optimum therefore exists. Other optima need not be sparse.

The sparse optimum has polynomial rational bit length. Clear the weight denominators by their product Q. Its bit length is at most the sum of the input denominator lengths. Each $Q\pi_e$ is an integer in $[0, Q]$. For a nonsingular n-column basis A, Cramer's rule gives a basic value with denominator $Q \det A$. Hadamard's bound gives $1 \le |\det A| \le n^{n/2}$ and bounds the numerator determinant by $Qn^{n/2}$. Thus each value has $O(\ell + n \log(n+1))$ bits. At most n nonzero entries and their indices have polynomial total length. When $n = 0$, the empty dual has cost zero.

A checker can verify a claimed rational upper bound h without finding every edge. It recomputes one-event feasibility for all positive-weight pairs. It rejects an omitted possible event, a listed impossible event, a self-edge, a negative entry, or an entry outside V_B. It combines repeated entries. For each edge with $\beta_{ef} > 0$, it uses (S29) to prove conflict. It checks every covering inequality. It also checks that the dual cost is at most h. All these checks use exact rational arithmetic. For any feasible correction, multiply each covering inequality by its event indicator and sum. Each vertex indicator is at most one. The sum of the two indicators on each checked edge is at most one. Therefore $C_{\pi,t}(z) \le \sum_e \alpha_e + \sum_{ef} \beta_{ef} \le h$. Edges with zero β need no check for this claimed upper bound. A dual certificate alone does not prove that $h = H_B(t)$. To establish this equality, also supply a feasible primal solution for the full conflict graph (V_B, F_B) with objective value h, or solve (S30) exactly.

S4.3 Supplied clique constraints

A clique is a set of events in which every two distinct members conflict. At most one of its events can occur under a correction. For any supplied family $\mathcal{C}$ of verified cliques, add $\sum_{e \in C} v_e \le 1$ for each $C \in \mathcal{C}$. Its optimum satisfies $U_B(t) \le H_B^{\mathcal{C}}(t) \le H_B(t)$. These are standard conflict-graph inequalities [S2]. If the total encoded family size is polynomial, all required edges can be checked

in polynomial bit complexity. The augmented rational program can also be solved within this complexity.

The dual adds nonnegative amounts γ_C to the objective and uses

$$\alpha_e + \sum_{f:\{e,f\}\in F_B} \beta_{ef} + \sum_{C\ni e} \gamma_C \geq \pi_e.$$

Multiplication by each feasible event indicator proves the upper bound: each vertex, edge, and clique contributes at most its coefficient. The sparse-basis and determinant argument above applies to the augmented matrix, whose entries remain in $\{-1, 0, 1\}$. No enumeration of all maximal cliques is required. If a separate family is chosen at each budget, the resulting bounds need not be nondecreasing. The breakdown bound below uses the original edge relaxation.

S4.4 The breakdown budget

If $B \leq B'$, then $Z_B \subseteq Z_{B'}$ and $V_B \subseteq V_{B'}$. An edge between old vertices in $F_{B'}$ already belongs to F_B. Extend any feasible vector for (S30) at B by zero on new vertices. Edge constraints between old vertices still hold. A constraint incident to a new vertex also holds because its new coordinate is zero and every coordinate is at most one. The objective is unchanged. Thus $H_B(t)$ is nondecreasing in B. At $B = 0$, the recorded correction is unique. Every possible event occurs there, so F_0 is empty and $H_0(t) = U_0(t)$.

For a rational reversal-weight limit $0 \leq \tau < 1$, let

$$\begin{aligned} b_* &= \min\{b \in \{0, \ldots, r\} : U_b(t) > \tau\}, \\ b_L &= \min\{b \in \{0, \ldots, r\} : H_b(t) > \tau\}. \end{aligned} \tag{S33}$$

If either set is empty, its minimum is defined to be $r + 1$. For every $b < b_L$, (S31) gives $U_b(t) \leq H_b(t) \leq \tau$. Therefore $b_L \leq b_*$. If $b_L = r + 1$, all permitted budgets are certified and $b_* = r + 1$. A feasible correction with exactly c changed labels and $C_{\pi,t}(z) > \tau$ belongs to Z_c and proves $b_* \leq c$, including $c = 0$. Hence a permitted correction whose reversal weight exceeds the limit gives $b_L \leq b_* \leq c$. Computing all $r + 1$ graph bounds has polynomial bit cost. Because the bounds are nondecreasing, binary search can also be used. A value $H_b(t) > \tau$ alone does not prove that a damaging correction exists.

S5 How to verify a supplied upper bound

Evaluating a feasible list gives a lower bound. An upper bound requires proof that the selected construction covers every correction. The following finite records are sufficient to check that coverage.[5]

Each record identifies the complete input, including predictions, fixed and recorded labels, flagged locations, error budget, order, comparison weights, original denominator, and tolerances. Check all rational values exactly. Recompute Σ and ν, and verify the construction selection in Section IV-D of the main paper: after the zero and recorded-label cases, choose the spectral construction when $r \leq B\lceil \log_2(r + 1)\rceil$ and the sample-mean construction otherwise. Check every output's binary labels and error budget. Record which output belongs to each construction item. Keep this association when outputs are duplicates.

[5] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

For a case that returns only the zero vector, verify its condition. For the direct recorded-label branch, check $\delta \geq 2B/N$ and the unchanged output. For the spectral small-energy branch, check $4Br/N^2 \leq \nu^2$. For an exact fallback, reconstruct every traversal node and include every required feasible vector. The spectral fallback scans the full cube and filters by the error budget. The sample fallback uses increasing subsets of size at most B. If the traversal is incomplete, coverage is unverified.

For a spectral cover, verify the stored rational subspace and projector. Check the residual inequality, score factor inequalities, shape, and mesh from Section S2. Repeating the exact construction is sufficient. Reconstruct every attempted child below every accepted prefix using the stated finite range and exact prefix inequality. Include rejected children and accepted prefixes without a full continuation. Check every accepted full cell. For each cell, supply either a rational feasible representative in K_B with its cell association, or a valid rational Farkas certificate that proves the cell is empty. For each nonempty cell, check its associated feasible output and squared rounding error at most $8B/N^2$. For rank zero, check the single formal cell. These checks establish the complete cell cover and error bound.

For a sample cover, reconstruct all $(r+1)^k$ ordered words and their exact means. For every word, replay all T conditional-gradient steps, including the tie rule, exact linear oracle, and step size. Check the final point, its associated feasible output, and the same rounding inequality. The convergence proof in Section S3 then applies to every mean. The checker need not solve the quadratic program exactly. It need not identify which word covers a particular correction. It must verify the steps and associated output for every required word.

The coverage records can be as large as the construction. They need not be short. The conflict dual in Section S4 has polynomial total bit length. Verify its event-oracle results and rational dual inequalities as specified there. The input error budget needs separate validation for an actual benchmark.

S6 Exact decisions, extensions, and examples

S6.1 Proof of the exact decision result

This subsection proves Proposition 4 of the main paper: deciding $U_B(t) > \tau$ is NP-complete, and deciding $U_B(t) \leq \tau$ is coNP-complete. Both statements hold with $B = r = N$, $t = \tau = 1/2$, and equal positive weights on a matching of model rows; all other comparisons have weight zero. The reference order may sort models by nonincreasing recorded accuracy. The number of models is part of the input, and repeated prediction rows are allowed. Hamming distance is the number of coordinates at which two strings differ.[6]

Proof. A corrected label vector is a witness of polynomial length for $U_B(t) > \tau$. Its budget and correct counts can be checked directly. The strict reversal events and their rational weighted sum can also be checked. These checks use a polynomial number of bit operations. Thus the violation problem belongs to NP, and its complement belongs to coNP.

We reduce binary Closest String, which is NP-complete [S4]. The input consists of strings $s_1, \ldots, s_n \in \{0,1\}^L$ and an integer radius $0 \leq d \leq L$, where $L \geq 1$. It asks whether a string x satisfies $d_H(x, s_i) \leq d$ for every i, where d_H is Hamming distance. For radii outside this range, the answer is immediate. Thus this restriction preserves NP-completeness.

[6] AI assistance in this section: [S1, S8, S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

Set

$$N = 4L + 4, \qquad q = 3L + 4, \qquad C = L + 2d + 3.$$

There are L original columns and q padding columns. Since $0 \le d \le L$, we have $L < C < q$. Split the padding into C discriminating columns and $q - C$ neutral columns. For each i, create two rows A_i, B_i. On the original columns, they predict $\overline{s_i}, s_i$, respectively. On each discriminating padding column, their predictions are $1, 0$. On each neutral padding column, both predict zero.

Add a pair A_0, B_0 that agrees at zero on the original columns and predicts $1, 0$ on every padding column. Add n more pairs whose rows are identical and predict zero everywhere. Give each of these $2n + 1$ designated pairs weight $1/(2n + 1)$, and give all other comparisons weight zero. Thus $M = 4n + 2$. Flag every column and set $B = r = N$.

Record zero on the original columns and one on all padding columns. Writing $D_i^0 = d_H(0, s_i)$, the recorded correct counts satisfy

$$c_{A_i} = C + D_i^0 > L \ge c_{B_i} = L - D_i^0.$$

The extra pair has recorded counts N, L. Each identical pair has counts L, L. Sort all rows by nonincreasing recorded count and resolve ties to orient each identical pair as designated. Every positive-weight pair then has its designated A row earlier than its B row. This gives the required recorded-accuracy reference order.

For corrected original labels x, let h be the number of ones on the discriminating padding columns. Active pair i has count difference

$$\Delta_i = L - 2d_H(x, s_i) + C - 2h, \qquad \Delta_i - N/2 = 2d + 1 - 2\big(d_H(x, s_i) + h\big).$$

Therefore its strict reversal at $t = 1/2$ occurs exactly when $d_H(x, s_i) + h \le d$. Changing a padding label from one to zero cannot decrease any designated pair's later-minus-earlier count difference. The full budget permits every such change. Thus there is a maximizer with all padding labels zero. At this maximizer, active pair i reverses exactly when $d_H(x, s_i) \le d$. The extra pair reverses because its count difference is $q > N/2$. Each identical pair has difference zero. Consequently,

$$U_N(1/2) = \frac{1 + \max_x |\{i : d_H(x, s_i) \le d\}|}{2n + 1}.$$

This exceeds $1/2$ exactly when all n strings are covered. In a no-instance it is at most $n/(2n+1)$; in a yes-instance it is $(n+1)/(2n+1)$. The construction has polynomial explicit size and polynomial rational bit length. This proves NP-completeness of the violation problem. Taking complements proves coNP-completeness of the certification problem. □

The proposition concerns exact decisions. The reduction uses zero weights outside the matching. It also uses padding columns and pairs of identical rows. It does not prove hardness for uniform weights on all model pairs. Its weight gap is $1/(2n + 1)$, and its active score gap at the margin is $1/N$. Thus the result does not establish hardness at fixed positive approximation tolerances. It also gives no lower bound that matches the exponent in Theorem 1 of the main paper.

One fixed replacement label. The list results also apply to categorical predictions when each flagged example has one fixed candidate replacement label. For flagged column u_k, let z'_k be its fixed replacement label, distinct from the recorded label $\widehat{z}_{u_k}$. The binary variable x_k selects this replacement. Thus X_B and K_B remain unchanged. For $e = (j, i)$, define

$$g_{eu}(v) = \mathbf{1}\{P_{iu} = v\} - \mathbf{1}\{P_{ju} = v\}, \qquad a_{ek} = g_{e,u_k}(z'_k) - g_{e,u_k}(\widehat{z}_{u_k}).$$

Each coefficient belongs to $\{-2,-1,0,1,2\}$. The score remains affine in x, and the bounds $\Sigma_{uu} \le 4$ and $\operatorname{tr}\Sigma \le 4r$ still hold. The rounding proof, both cover constructions, the recorded-label branch, and the threshold transfer use these bounds and the sets X_B, K_B. These proofs do not require even coefficients. Their guarantees and constants therefore remain valid.

The conflict dynamic program also permits any explicitly listed finite label domain $\mathcal{D}_u$ that contains the recorded label. Each transition still adds two contributions in $\{-1,0,1\}$ and a change cost in $\{0,1\}$. Its state proof is unchanged, with $D = \sum_u |\mathcal{D}_u|$ replacing the binary bound $D \le 2N$. The two-event transition count remains $O(N^2(B+1)D)$. This broader domain statement concerns the conflict certificate. The list extension above permits only one fixed replacement per example.

S6.2 Two construction regimes

Put $D_0 = \epsilon\delta^2 N^2$. The following two families compare the stated sufficient bounds. In each family, the inputs vary to show how those bounds grow. For integer $h \ge 2$, set $N = r = h^3$, $B = h^2$, $\delta = 1/h$, and $\epsilon = 1/4$. The spectral sufficient factor is $2^{O(h)}$. The sample factor is polynomial in r, because $B^2/D_0 = 4$. In fact, $\nu = 1/(2h)$ gives $k = 256$ and $T = 2048h$; the large-pool branch applies for $h \ge 23$. The full-budget spectral factor is $2^{O(h^2)}$, and complete feasible enumeration has log-size $\Theta(h^2 \log h)$. Indeed, with $r = h^3$ and $B = h^2$, its size is at least $\binom{r}{B} \ge (r/B)^B = h^{h^2}$ and at most $(B+1)r^B$. Thus this family gives a polynomial sample-mean bound, which is smaller than the stated spectral bound. The polynomial degree can still be large.

The complementary family has $N = r = 2h^3$, $B = h^3$, $\delta = 1/h$, and $\epsilon = 1/4$. Here $0 < B < r$, $Br/D_0 = 2h^2$, and $B^2/D_0 = h^2$. The spectral exponent is $O(h^2)$, whereas the sample exponent is $O(h^2 \log h)$. Complete feasible enumeration has log-size $\Theta(h^3)$: with $r = 2B$, symmetry gives $2^{r-1} \le \sum_{j=0}^{B} \binom{r}{j} \le 2^r$. The recorded-label singleton condition fails in both families.

These families compare sufficient upper bounds. They do not give a lower bound on construction cost. They also do not compare actual runtime with enumeration or the conflict-aware certificate. Further analysis is needed for lists with several candidate replacements per example, unequal correction costs, or several simultaneous error budgets.

S6.3 A restriction missed by pair conflicts

Two-event tests can miss a restriction on three or more events. Take $N = r = B = 3$, so every binary label assignment is permitted regardless of the recorded labels. Use centers $000, 110, 101, 011$. For each center, make a pair of model rows. The later row predicts the center, and the earlier row predicts its complement. Order these four row pairs consecutively and give each within-pair comparison weight $1/4$. All other pairs have weight zero. At margin $t = 1/6$, the score is $1 - 2d/3$, where d is Hamming distance from the center, the number of labels that differ. The reversal event is exactly the radius-one ball. It contains the center and each word that differs from it in one label. Every two balls meet, so the graph has no edges and $H_3 = 1$. A word with even parity has an even number of ones. It belongs to one ball. A word with odd parity has an odd number of ones. It belongs to three balls. These are all eight words, so $U_3 = 3/4$. A bound above the reversal-weight limit can therefore remain unresolved.

S7 Exploratory synthetic implementation check

This check tests the implemented binary event tests against exhaustive correction enumeration. It also compares the conflict-aware breakdown lower bound with separate tests and the exact

breakdown budget. This is an exploratory check. Its configuration was fixed before generating the 20 synthetic inputs analyzed here. All results reported in this section use these inputs.[7]

Synthetic inputs and audit rule. For each of 20 seeds, $2026103000, \ldots, 2026103019$, initialize NumPy's `default_rng` separately. Use $N = 200$ examples, $M = 6$ models, and $r = 12$ flagged labels. First draw binary true labels Y_u independently with probability $1/2$ for each label. Next draw independent example difficulties $D_u \sim \mathcal{N}(0, 1.5^2)$ and model skills $S_m \sim \mathcal{N}(2, 0.15^2)$. Draw independent $V_{mu} \sim \mathrm{Uniform}(0, 1)$ and set

$$P_{mu} = \begin{cases} Y_u, & V_{mu} < \big(1 + \exp(-(S_m - D_u))\big)^{-1}, \\ 1 - Y_u, & \text{otherwise.} \end{cases}$$

Then flip each true label independently with probability 0.03 to obtain its recorded label $\widehat{Y}_u$. Flag the 12 examples with the largest values of $\sum_m \mathbf{1}\{P_{mu} \neq \widehat{Y}_u\}$, with increasing example index for ties. These counts measure disagreement with the recorded labels. Sort models by nonincreasing recorded accuracy, with increasing model index for ties. Give all $Q = \binom{6}{2} = 15$ reference-ordered pairs weight $1/15$. Use margin $t = 0$ and limits $\tau = 1/10$ and $1/5$. Only strict corrected-score reversals count. The saved environment records Python 3.12.14 and NumPy 2.3.5.

Exact computation and verification. Enumerate all $2^{12} = 4096$ correction subsets for each input. Recompute correct counts after flipping the selected recorded labels. At each budget $b = 0, \ldots, 12$, maximize the number of simultaneous reversals among subsets of size at most b. Divide this maximum by $Q = 15$ to obtain $U_b(0)$ exactly. Compare the implemented one-event and two-event tests with this enumeration at every budget.

For each budget b, let $V = V_b$ be the vertex set of the resulting conflict graph. Solve its uniform-weight edge program exactly through the bipartite double cover, which gives each vertex a left copy and a right copy. Each conflict edge ef gives edges $e_L f_R$ and $f_L e_R$. Compute a maximum matching and a vertex cover of the same size. If c_{v_L}, c_{v_R} indicate cover membership, then

$$x_v = 1 - \frac{c_{v_L} + c_{v_R}}{2}$$

is feasible for the unweighted edge program. Give each conflict edge half the number of its two orientations used by the matching. Give each vertex a dual amount equal to one minus its incident edge amounts. Matching disjointness makes these amounts nonnegative. The primal objective is $|V| - |C|/2$, and the dual objective is $|V| - |\mathcal{M}|/2$, where C is the cover and $\mathcal{M}$ is the matching. Equal cover and matching sizes give equal feasible objectives. Dividing all objective weights and dual amounts by Q proves the exact value of $H_b(0)$. All coordinates are half-integral before this normalization. Integer checks verify feasibility, matching and cover conditions, and objective equality without numerical tolerances.

A separate scalar enumeration pass reads the saved inputs, flips labels, and recounts every prediction. It does not use the gain formulas or the first NumPy enumeration. It independently recomputes the minimum numbers of flips needed to realize each event and each event pair, the exact objective values, and their witnesses. The pass also checks every saved graph certificate and the reported budgets b_{sep} (defined below), b_L, and b_*. Each planned input was run once, followed by one verification pass over the saved inputs. Fixed wall-time limits were 600 seconds for the run, 30 per input, 5 per graph certificate, and 300 for verification. No planned input was replaced, and no additional runs or parameter changes were used to obtain the reported results.

[7] AI assistance in this section: [S9]. The AI assistance disclosure in this supplement describes the scope of assistance.

Complete sample and outcomes. All 20 planned inputs completed and passed the independent saved-input check. No input failed, timed out, remained incomplete, or was unattempted. The sample contains 81,920 enumerated corrections, 3,900 one-event budget checks, 27,300 two-event budget checks, and 260 exact graph certificates. There were no event-test mismatches or certificate failures. Define the separate-event breakdown bound by

$$b_{\mathrm{sep}} = \min\{b \in \{0, \dots, r\} : B_{\mathrm{sep},b}(0) > \tau\},$$

with an empty-set minimum of $r + 1$. Table S1 reports all 20 inputs for each limit. The value $r + 1 = 13$ denotes no violation through all r flagged changes. It is not a finite breakdown budget. In every no-violation case, $b_L = 13$ also certifies all 12 permitted changes.

Table S1: Complete exploratory sample: 20 planned, 20 complete, and 20 independently checked inputs; zero failed, timed-out, incomplete, or unattempted inputs. Finite equality counts exclude $b_* = r + 1$.

τ	$b_* \le r$	$b_* = r + 1$	$b_L > b_{\mathrm{sep}}$	$b_L = b_*$, finite
1/10	16	4	5/20	16/16
1/5	5	15	10/20	3/5

Code, data, and limits. The accompanying reproducibility materials contain the code, fixed configuration, synthetic inputs, results for every budget, rational certificates, completion and environment records, verification records, file manifests, and reproduction instructions. This finite check does not prove general software correctness. It does not measure a runtime advantage over exact solvers or performance on real prediction tables. It does not estimate how often the bound helps in larger or different samples. The list constructions were not implemented or evaluated in this check.

AI assistance disclosure

OpenAI Codex, with GPT-6 Sol, GPT-6 Astra, and GPT-6.1 Sol, assisted with proof development, mathematical and scientific checks, drafting, and revision. OpenAI assistance with the synthetic implementation check covered code design, execution, verification, and results wording. The affected text includes the abstract, impact statement, all main sections, and all supplement sections. ChatGPT suggested the sample-mean comparison and the two-example illustration. Anthropic Claude Opus 5.5 also provided scientific assistance with the mathematical arguments and manuscript. The assistance included mathematical content and was not limited to language editing. The author is responsible for the claims and text.

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